

# Machine Learning:

## Linear Regression

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# Goals

- Understand what linear regression is and why we use it
- Learn the linear regression model
- Understand least squares estimation
- Derive the closed-form solution for the slope and intercept
- Understand how to interpret linear regression coefficients

# Why study linear regression?

- In statistics, *regression analysis* includes any techniques for establishing relationships between a *dependent variable* and one or more *independent variables*.
- Linear regression is one of the most important tools in statistics, data science, and machine learning.

# Why study linear regression?

- Applications:
  - Biology: dose vs response
  - Medicine: exposure vs health outcome
  - Engineering: input vs output
  - Social science: study time vs exam score
- Key roles
  - Prediction
  - Explanation
  - Inference



# What is regression?

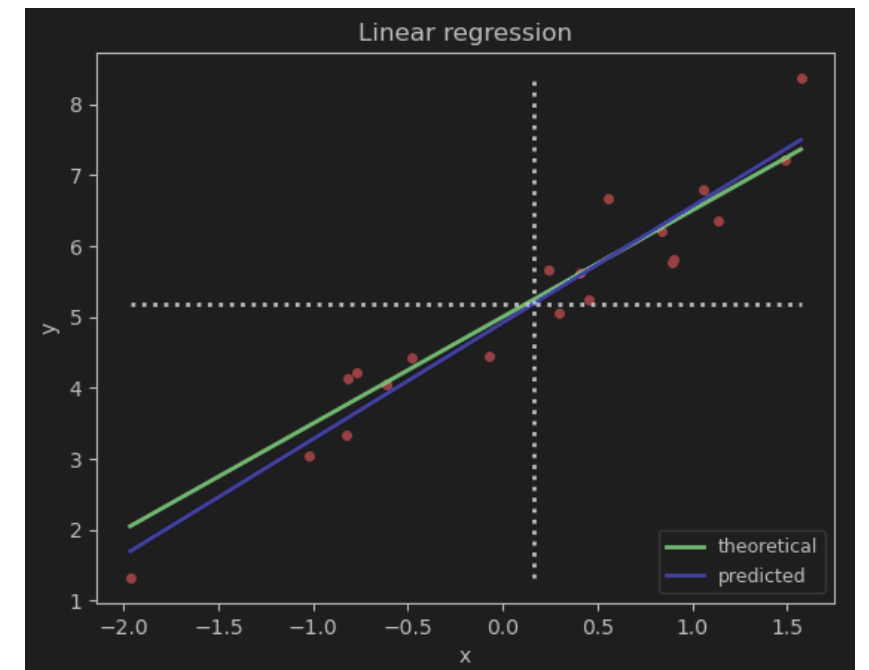
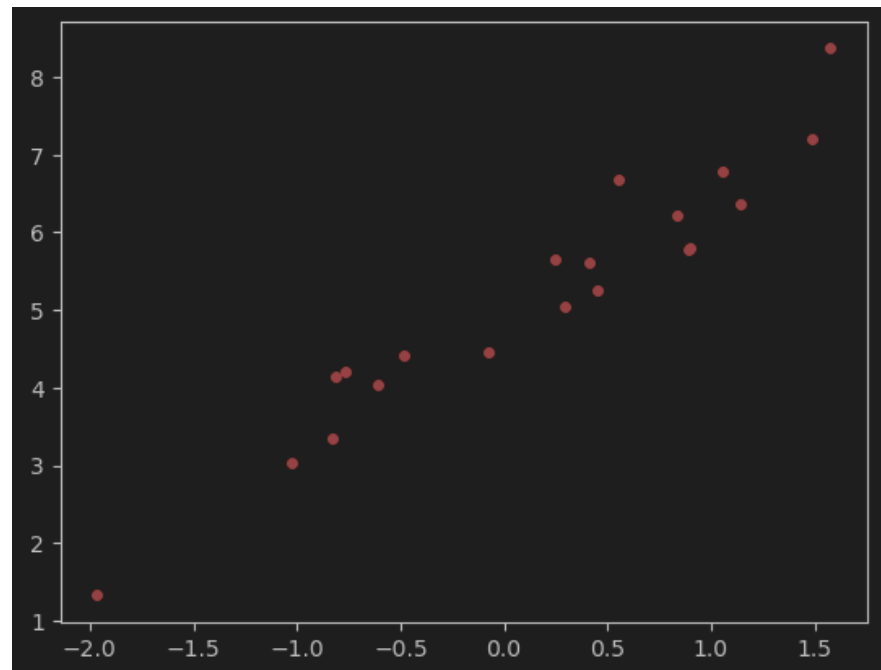
- Regression studies the relationship between:
  - A response variable (outcome):  $y$
  - One or more predictors (inputs):  $x$
- Goal:
  - Describe how  $y$  changes as  $x$  changes
  - Predict  $y$  from  $x$
- Key idea: Find a mathematical function that best summarizes the relationship between  $x$  and  $y$

# A First Look?

Suppose we observe data points:  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  Plot:

- $x$  on the horizontal axis
- $y$  on the vertical axis

Question: Can we summarize this cloud of points with a straight line?



# Simple Linear Regression

# Simple Linear Regression Model

- When there is only one independent variable  $x$ , we are doing a simple linear regression (SLR).
- Given observation pairs  $(x_i, y_i), i = 1, 2, \dots, n$ , we can use a model (affine function) to completely describe the distribution that generated the data. The model can be described as

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where:

- $\beta_0$  is the *intercept*
- $\beta_1$  is the *slope*
- $\epsilon_i$  is the random error.

# Interpreting the Parameters

- Slope ( $\beta_1$ )
  - Expected change in  $y$  when  $x$  increases by 1 unit
  - Direction and strength of relationship
- Intercept ( $\beta_0$ )
  - Predicted value of  $y$  when  $x = 0$
  - May or may not have a practical interpretation
- Error term ( $\epsilon_i$ )
  - Captures measurement error, randomness, unobserved factors and follows a normal distribution with mean 0 and standard deviation  $\sigma$ , i.e.

$$\epsilon \sim \mathcal{N}(0, \sigma^2 \cdot I_{n \times n})$$

# What does ``Best Line'' Mean?

Many lines could fit the same data. We need a rule to decide:

- Which line is the best?

Idea:

- Compare predicted values to observed values
- Measure how far the points are from the line

This leads to residuals.

# Residuals

For each observation:

$$r_i = y_i - \hat{y}_i$$

Where:

- $y_i$  = observed value
- $\hat{y}_i$  = predicted value from the line

Interpretation:

- Residual = vertical distance from point to fitted line



# Simple Linear Regression

- In the model, we have

$$E[Y|X = x] = \beta_0 + \beta_1 x,$$

i.e. the conditional expectation of  $Y$  given  $X$  is linear in  $X$ . Hence, it is a *linear* regression model. We assume that the distribution is homoscedastic, so that  $\sigma(Y|X = x) = \sigma$ .

- For fixed values of  $\beta_0$  and  $\beta_1$  (not necessarily the true values),  $r_i = y_i - \beta_0 - \beta_1 x_i$  is called the *residual* at  $x_i$ . Note that  $r_i$  is the vertical distance from  $y_i$  to the line  $\beta_0 + \beta_1 x$ .
- We have reduced the problem to three unknowns (parameters):  $\beta_0, \beta_1$  and  $\sigma$ . Now we need a way to estimate these unknowns from the data.



# Least Squares Solution

- One common estimation technique for linear regression is *ordinary least squares* (OLS). That is, we will need to find  $\beta_0$  and  $\beta_1$  that would minimize the sum of squared residuals of the linear regression model.
- In other words,  $\beta_0$  and  $\beta_1$  solve the following minimization problem:

$$(\hat{\beta}_0, \hat{\beta}_1) = \operatorname{argmin}_{(\beta_0, \beta_1)} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

When  $\beta_0$  and  $\beta_1$  are chosen so the fit to the data is good, the sum of squared residuals will be small. If  $\beta_0$  and  $\beta_1$  are chosen so the fit to the data is poor, the sum of squared residuals will be large.

# Least Squares Solution

- Denote  $SS_{err}(\beta_0, \beta_1)$  as the sum of squared error:

$$SS_{err}(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

Then the optimal  $\beta_0$  and  $\beta_1$ , in the sense of least squares, can be obtained by solving the following equations:

$$\frac{\partial SS_{err}}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)$$

$$\frac{\partial SS_{err}}{\partial \beta_1} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) \cdot x_i$$

# Least Squares Solution

- The solution to the above two simultaneous equations are

$$\hat{\beta}_1 = \frac{S_{yx}}{S_{xx}}$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where

$$S_{yx} = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})$$
$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

# Least Squares Solution

- With some simple manipulation of  $\hat{\beta}_1$ , we can easily see that  $\hat{\beta}_1$  can be written as

$$\hat{\beta}_1 = r_{xy} \frac{S_{yy}}{S_{xx}} = \frac{\text{cov}(X, Y)}{\text{var}(X)}$$

where  $r_{xy}$  is the *sample correlation coefficient* between  $X$  and  $Y$ .  $S_{xx}$  and  $S_{yy}$  are the *sample variance* for  $X$  and  $Y$ , respectively.

$$\begin{aligned} r_{x,y} &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \\ &= \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n \sum x_i^2 - (\sum x_i)^2} \sqrt{n \sum y_i^2 - (\sum y_i)^2}} \end{aligned}$$

$$S_{xx} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

$$S_{yy} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2}$$

# Least Squares Solution

- Therefore, the OLS coefficient estimates are closely related to the correlation coefficient between the covariate and the response.
- So the estimated regression function is

$$\hat{E}[Y|X = x] = \hat{\beta}_0 + \hat{\beta}_1 x$$

and the *fitted values* are

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

# Least Squares Solution

Interpretation of:

$$\bar{\beta}_1 = \frac{cov(X, Y)}{var(X)}$$

- If  $x$  barely varies  $\rightarrow$  denominator is tiny  $\rightarrow$  slope becomes unstable.
- if  $x$  varies a lot  $\rightarrow$  slope estimate is more stable. We cannot estimate a slope reliably if the predictor barely changes.

# Fitted Values

With  $\hat{\beta}_0$  and  $\hat{\beta}_1$ , we can estimate  $\hat{y}_i$  as:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

This gives:

- Predicted value for each  $x$
- The fitted regression line

Residuals:

$$r_i = y_i - \hat{y}_i$$



# Key Properties of Least Squares

- Some properties of the least squares estimates:

1).  $\hat{\beta}_1 = r_{xy} \frac{S_{yy}}{S_{xx}}$ , so  $\hat{\beta}_1$  and  $r_{xy}$  always have the same sign. If the data are positively correlated, the estimated slope is positive, and if the data are negatively correlated, the estimated slope is negative.

2). The fitted line  $\hat{\beta}_0 + \hat{\beta}_1 x$  always passes through the overall mean  $(\bar{x}, \bar{y})$ .

3). Since  $\text{cov}(cX, Y) = c \cdot \text{cov}(X, Y)$  and  $\text{var}(cX) = c^2 \cdot \text{var}(X)$ , if we scale  $X$  values by  $c$ , then the slope is scaled by  $1/c$ . If we scale the  $Y$  values by  $c$ , then the slope is scaled by  $c$ .

- Once we have  $\hat{\beta}_0$  and  $\hat{\beta}_1$ , we can compute  $r_i$  and use  $r_i$  to estimate  $\sigma$ :

$$\hat{\sigma} = \sqrt{\frac{\sum_i r_i^2}{n - 2}}$$



# Key Properties of Least Squares

The residual variance  $\hat{\sigma}^2$  tells us:

- How much scatter there is around the regression line.
- How well the line explains the data.

Small  $\hat{\sigma}^2$ :

- Data points tightly follow the line.
- Strong linear relationship.

Large  $\hat{\sigma}^2$ :

- Lots of unexplained noise.
- Weak linear relationship

# Prediction vs Inference

Linear regression can be used for two purposes: Prediction and Inference.

## Prediction

- Goal: predict future or unseen outcomes
- Question: How accurate are our predictions?
- Focus:
  - Test error: MSE (Mean Squared Error), RMSE (Root Mean Squared Error), where

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

and

$$RMSE = \sqrt{MSE}$$

- Generalization to new data

# Prediction vs Inference

Use linear regression for inference

- Goal: understand relationships between variables
- Question: What is the effect of  $x$  on  $y$ ? Is it significant?
- Focus
  - Coefficients
  - Standard errors
  - p-values and confidence intervals
- Interpretability: essential
- Assumptions matter greatly

Prediction asks “What will happen?”

Inference asks “What is the relationship, and how strong is it?”

# Assumptions

For linear regression to work well, we assume:

- Linearity: mean of  $y$  is linear in  $x$ , i.e.

$$E[Y|X = x] = \beta_0 + \beta_1 x$$

- Independence of observations: Each observation is collected independently and does not influence the others.
- Constant variance (homoskedasticity): The spread of  $y$  around the fitted line does not depend on  $x$ .
- Mean of errors is zero:  $E[r_i|x_i] = 0$

# Multiple Linear Regression

# Multiple Linear Regression

So far, we used one predictor  $x$ . In many real problems, we use several predictors:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_d x_d + \epsilon$$

where

- $x_1, x_2, \dots, x_d$  are predictors.
- Each coefficient measures a separate effect.

Examples

- House price  $\sim$  size + location + age
- Exam score  $\sim$  study time + GPA + attendance
- Health outcome  $\sim$  exposure + age + sex

Key idea: We model  $y$  using several variables at the same time.

# Interpreting Coefficients

In MLR:

$\beta_i$  = effect of  $x_i$  on  $y$  holding all other variables fixed

Interpretation:

- Change in  $y$  for a 1-unit increase in  $x_i$
- While keeping all other predictors constant
- Coefficients represent adjusted effects.



# Interpreting Coefficients

Why multiple regression is needed:

- Real variables are correlated
- Simple regression can be misleading

Example:

- Salary vs education
- But experience also affects salary

Multiple regression allows us to:

- Adjust for confounders
- Separate effects
- Make fair comparisons

Key idea: Multiple regression isolates the effect of one variable while controlling for others.



# Design Matrix

Matrix notation

$$Y = X\beta + \epsilon$$

Where  $X$  is the design matrix with the first column being 1's (intercept) and the other columns being predictors.

OLS solutions:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Interpretation:

- Same least squares idea
- Works for any number of predictors.

In python, regression is always done using this matrix form.

# Summary

So far, we have learned:

- What regression is and why we use it
- The simple linear regression model
- The idea of residuals
- Least squares estimation
- Closed-form formulas for intercept and coefficients

Key idea: Linear regression finds the line that minimizes squared prediction errors

# Interpreting Coefficients

# Interpreting Coefficients

- For a linear regression model:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_d x_d$$

- Interpretation of  $\beta_j$  for  $j = 1, 2, \dots, d$ :
  - If  $x_j$  increases by 1 unit, then the predicted  $y$  changes by  $\beta_j$  units, holding all other predictors constant.

# Interpreting Coefficients: Example 1

Example (single predictor)

- Suppose:

$$\hat{y} = 50 + 2x$$

where  $x$  = hours studied and  $y$  = exam score.

- Interpretation
  - Intercept  $\beta_0 = 50$ : predicted exam score when study hours = 0 is 50.
  - Slope  $\beta_1 = 2$ : each extra hour studied increases predicted score by 2 points.

# Interpreting Coefficients: Example 2

Example (multiple predictors)

- Suppose:

$$\hat{y} = 30 + 2x_1 + 10x_2$$

where:

- $x_1$  = hours studied
- $x_2$  = tutoring (0 = no, 1 = yes)
- $y$  = exam score.

- Interpretation

- $\beta_0 = 30$ : predicted exam score when study hours = 0 is 30.
- $\beta_1 = 2$ : for two students with the same tutoring status, the one who studied 1 extra hour is predicted to score 2 points higher.
- $\beta_2 = 10$ : for two students with the same study hours, the one who got tutoring is predicted to score 10 points higher.

# Interpreting Coefficients

- The intercept is:

$$\beta_0 = \text{predicted } y \text{ when all } x = 0$$

That is only meaningful if “all predictors = 0” is realistic.

- Example:
  - if  $x_1$  is “age” and  $x_2$  is “income”, then age=0 is nonsense.
  - So the intercept may be mathematically required but not practically interpretable.



# Interpreting Coefficients

If predictors are scaled/transformed

- If you standardize  $x$ :

$$x^* = \frac{x - \mu}{\sigma}$$

- Then  $\beta$  means:
  - A 1 standard deviation increase in  $x$  changes  $y$  by  $\beta$  units.



# Interpreting Coefficients

Categorical variables (dummy variables)

- If you encode:
  - Sex = Male or Female
  - Female = 0 and male = 1

- Then

$$\hat{y} = \beta_0 + \beta_1 \cdot \text{Male}$$

- Interpretation:
  - $\beta_0$ : predicted mean for females
  - $\beta_1$ : difference between males and females (male - female)

# Interpreting Coefficients

The most important warning:

**Coefficients are NOT necessarily causal.**

Even if  $\beta_1$  is large and significant, it could be due to:

- confounding
- omitted variables
- reverse causation
- measurement bias.

Linear regression is fundamentally **association**, unless you have a causal design.

# Interpreting Coefficients

Quick summary:

- $\beta_j$ : effect of  $x_j$  on  $y$ , holding other predictors fixed.
- **Sign**: direction of relationship (+ or -)
- **Magnitude**: how much  $y$  changes per 1 unit change in  $x_j$
- **Intercept**: predicted  $y$  when all predictors = 0 (sometimes meaningless).

Linear regression is fundamentally **association**, unless you have a causal design.

# Confounding: Example 1

Example of confounding

We want to model:

- $y$  = exam score
- $x_1$  = ability (0 = low, 1 = high)
- $x_2$  = hours studied

| Student | Ability $x_1$ | Study Hours $x_2$ | Score $y$ |
|---------|---------------|-------------------|-----------|
| 1       | 0             | 1                 | 52        |
| 2       | 0             | 2                 | 54        |
| 3       | 0             | 3                 | 56        |
| 4       | 1             | 4                 | 82        |
| 5       | 1             | 5                 | 84        |
| 6       | 1             | 6                 | 86        |

# Confounding: Example 1

Example of confounding: Fit the WRONG model (omit ability  $x_1$ )

- Model A:

$$y = \beta_0 + \beta_2 x_2$$

- Learning  $\beta_0$  and  $\beta_1$  from the data, we get:

$$\hat{y} = 40.40 + 8.17x_2$$

- Interpretation (WRONG but what the models tells you):
  - Each extra hour studied increases score by  $\sim 8.17$  points.

# Confounding: Example 1

Example of confounding: Fit the CORRECT model (control ability  $x_2$ )

- Model B:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

- Learning  $\beta_0$  and  $\beta_1$  from the data, we get:

$$\hat{y} = 50 + 24x_1 + 2x_2$$

- Interpretation (CORRECT):
  - $\beta_2 = 2$ : If two students have the same ability level, the one who studies 1 more hour is predicted to score 2 points higher.
  - $\beta_1 = 24$ : If two students study the same number of hours, the high-ability student scores 24 points higher on average.

# Confounding: Example 1

## Comparison

| Model             | Regression                                | $\beta_2$              |
|-------------------|-------------------------------------------|------------------------|
| Model A (wrong)   | $y = \beta_0 + \beta_2 x_2$               | $\beta_2 \approx 8.17$ |
| Model B (correct) | $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ | $\beta_2 = 2$          |

$\beta_2$  dropped from 8.17 to 2 once we controlled for ability. Why this happens?  
Ability is a confounder because:

- Ability  $\uparrow \rightarrow$  Study hours  $\uparrow$
- Ability  $\uparrow \rightarrow$  Exam score  $\uparrow$
- So when you omit ability, the model mistakenly thinks that “Study hours must be causing that big score jump.”



# Confounding: Example 2

Example: Simpson's paradox.

The dataset.

- Low ability group ( $x_1 = 0$ ): Studying helps (positive slope)

| Student | Ability $x_1$ | Study Hours $x_2$ | Score $y$ |
|---------|---------------|-------------------|-----------|
| 1       | 0             | 4                 | 60        |
| 2       | 0             | 5                 | 65        |
| 3       | 0             | 6                 | 70        |

Slope = +5 per hour

- High ability group ( $x_1 = 1$ ): Studying also helps (positive slope)

| Student | Ability $x_1$ | Study Hours $x_2$ | Score $y$ |
|---------|---------------|-------------------|-----------|
| 4       | 1             | 1                 | 90        |
| 5       | 1             | 2                 | 95        |
| 6       | 1             | 3                 | 100       |

Slope = +5 per hour



# Confounding: Example 2

Example: Simpson's paradox.

- Step 1: Within each group, the effect is positive.
  - For low ability: 4  $\rightarrow$  6 hours gives 60  $\rightarrow$  70 (upward trend)
  - For high ability: 1  $\rightarrow$  3 hours gives 90  $\rightarrow$  100 (upward trend)
  - So the true relationship within groups is:

more studying  $\Rightarrow$  higher score

# Confounding: Example 2

Example: Simpson's paradox.

- Step 2: But overall, it looks negative. Ignore ability  $x_1$  and just look at all 6 points:

| Study Hours $x_2$ | Score $y$ |
|-------------------|-----------|
| 1                 | 90        |
| 2                 | 95        |
| 3                 | 100       |
| 4                 | 60        |
| 5                 | 65        |
| 6                 | 70        |

Overall pattern:

- Small  $x_2$  values have high  $y$  values (90 - 100).
- Large  $x_2$  values have low  $y$  values (60 - 70).

So the overall regression line will slope downward.

# Confounding: Example 2

Example: Simpson's paradox.

- Step 3: Compute the “overall” slope

$$y = \beta_0 + \beta_1 x_2 = 103 - 6.57x_2$$

Interpretation (WRONG):

- Each extra hour studied lowers score by about 6 points.
- Step 4: The confounding story
  - High ability students score high even if they study less.
  - Low ability students score low even if they study more.
  - So when you mix the groups, it creates the illusion that “Studying more is associated with lower scores, even though within each group, studying helps.”

# Confounding: Example 2

Example: Simpson's paradox.

- Step 4: Compute the correct regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 = 40 + 45x_1 + 5x_2$$

- $\beta_1 > 0$ : high ability boosts score.
- $\beta_2 > 0$ : studying helps, within ability level.

- Summary

- Within each ability group:

study hours  $\uparrow \Rightarrow$  score  $\uparrow$

- Overall (ability omitted):

study hours  $\uparrow \Rightarrow$  score  $\downarrow$

This is a classic sign-flip due to confounding / Simpson's paradox.

# Inference

How confident are we in  $\hat{\beta}_0$ ,  $\hat{\beta}_1$ , and  $\hat{\sigma}$ ?

# Sum of Squares and Goodness of Fit

- We would like to understand how the following quantities are related:

1).  $y_i - \bar{y}$

2).  $y_i - \hat{y}_i = r_i$

3).  $\hat{y}_i - \bar{y}$

All three averages out to zero over the data.

- The residual  $r_i = y_i - \hat{y}_i$  and the fitted values  $\hat{y}_i$  are uncorrelated.

# Sum of Squares and Goodness of Fit

- With the sum of squares terminology in the Table below, we have,

$$SS_{tot} = SS_{reg} + SS_{err}.$$

Table 1.1: Sum of squares terminology.

| Formula                        | Name                      | Abbrev.    |
|--------------------------------|---------------------------|------------|
| $\sum (y_i - \bar{y})^2$       | Total sum of squares      | $SS_{tot}$ |
| $\sum (\hat{y}_i - \bar{y})^2$ | Regression sum of squares | $SS_{reg}$ |
| $\sum (y_i - \hat{y}_i)^2$     | Residual sum of squares   | $SS_{err}$ |



# Sum of Squares and Goodness of Fit

- Corresponding to each sum of squares is a degree of freedom. Dividing the sum of squares by the df gives the “mean square” as shown in Table below.

Note that the  $MS_{tot}$  is the sample variance, and the  $MS_{err}$  is the estimate of  $\hat{\sigma}^2$  in the regression model.

The  $SS$  values add:  $SS_{tot} = SS_{reg} + SS_{err}$  and the degrees of freedom add:  $n - 1 = (n - 2) + 1$ .

The  $MS$  values do NOT add:  $MS_{tot} \neq MS_{reg} + MS_{err}$ .

Table 1.2: MS terminology.

| Formula                              | df      | Abbrev.    |
|--------------------------------------|---------|------------|
| $\sum (y_i - \bar{y})^2 / (n - 1)$   | $n - 1$ | $MS_{tot}$ |
| $\sum (\hat{y}_i - \bar{y})^2$       | 1       | $MS_{reg}$ |
| $\sum (y_i - \hat{y}_i)^2 / (n - 2)$ | $n - 2$ | $MS_{err}$ |



# Sum of Squares and Goodness of Fit

- If the model fits the data well,  $MS_{reg}$  will be large and  $MS_{err}$  will be small. Conversely, if the model fits the data poorly then  $MS_{reg}$  will be small and  $MS_{err}$  will be large. Thus the statistic

$$F = \frac{MS_{reg}}{MS_{err}}$$

can be used to evaluate the fit of the linear model (bigger  $F$  = better fit).

The distribution of  $F$  is an  $F$  distribution with  $1, (n - 2)$  df, or  $F_{1,n-2}$ .

We can test the null hypothesis that the data follow a model  $y_i = \mu + \epsilon_i$  against the alternative that the data follow a model  $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$  using the  $F$  statistic (an “ $F$  test”). A computer package or a table of the  $F$  distribution can be used to determine a  $p$ -value.

# Sum of Squares and Goodness of Fit

- In the case of simple linear regression, the  $F$  test is equivalent to the hypothesis test  $\beta_1 = 0$  versus  $\beta_1 \neq 0$ . Later when we come to multiple linear regression, this will not be the case.

A useful way to think about what the  $F$ -test is evaluating is that the null hypothesis is all  $Y$  values have the same expected value and the alternative is that the expected value of  $y_i$  depends on the value of  $x_i$ .

- With the sum of squares law, we can define the *Coefficient of determination*  $R^2$  as follows

$$R^2 = \frac{SS_{reg}}{SS_{tot}} = 1 - \frac{SS_{err}}{SS_{tot}} = (\widehat{cor}(X, Y))^2.$$

$R^2$  tells us how much variability in the  $Y$ 's is explained by the regression. With least squares regression,  $R^2$  is also equal to the square of the correlation coefficient between the response and explanatory variables.

$R^2 = 1$  indicates that the fitted model explains all variability in  $Y$ , while  $R^2 = 0$  indicates no “linear relationship” between the explanatory and response variables. For straight line regression, this means that the straight line model is a constant line, that is, slope=0, intercept= $\bar{y}$ .

An interior value such as  $R^2 = 0.7$  may be interpreted as follows: approximately seventy percent of the variation in the response variable can be explained by the explanatory variable. The remaining thirty percent can be explained by unknown, lurking variables or inherent variability.

# Statistical Properties of the OLS Estimates

- Since  $\hat{\beta}_0$  and  $\hat{\beta}_1$  are functions of the data, which is random, they are random variables, and hence they have a distribution.

This distribution reflects the sampling variation that causes  $\hat{\beta}_0$  and  $\hat{\beta}_1$  to differ somewhat from the population values  $\beta_0$  and  $\beta_1$ .

The sampling variation is less if the sample size  $n$  is large, and if the error standard deviation  $\sigma$  is small.

The sampling variation of  $\hat{\beta}_1$  is less if the  $x_i$  values are more variable.

# Statistical Properties of the OLS Estimates

- Summary of the properties of the distribution of  $\hat{\beta}_0$  and  $\hat{\beta}_1$ :
  1. Both are unbiased.

$$E[\hat{\beta}_0] = \beta_0$$

$$E[\hat{\beta}_1] = \beta_1$$

2.

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{(n-1)\text{var}(X)}$$

$$\text{var}(\hat{\beta}_0) = \sigma^2 \frac{\sum x_i^2 / n}{(n-1)\text{var}(X)} = \text{var}(\hat{\beta}_1) \frac{\sum x_i^2}{n}$$

- These properties allow us to compute confidence intervals for  $\hat{\beta}_0$  and  $\hat{\beta}_1$  for a specified confidence level.

# Statistical Properties of the OLS Estimates

- Start with the basic inequality for standardized  $\hat{\beta}_1$ :

$$P \left( -1.96 \leq \hat{\sigma}_X \sqrt{n-1} \frac{\hat{\beta}_1 - \beta_1}{\sigma} \leq 1.96 \right) = 0.95$$

then get  $\beta_1$  alone in the middle

$$P \left( \hat{\beta}_1 - 1.96 \frac{\sigma}{\hat{\sigma}_X \sqrt{n-1}} \leq \beta_1 \leq \hat{\beta}_1 + 1.96 \frac{\sigma}{\hat{\sigma}_X \sqrt{n-1}} \right) = 0.95$$

Replace 1.96 with 1.64, etc. to get CI's with different coverage probabilities.

# Statistical Properties of the OLS Estimates

- Note that in general we will not know  $\sigma$ , so we will need to plug-in  $\hat{\sigma}$  for  $\sigma$ . This plug-in changes the sampling distribution to  $t_{n-2}$ , so to be exact, we would replace the 1.96 in the above formula with  $Q_T(.975)$ , where  $Q_T$  is the quantile function of the  $t_{n-2}$  distribution.

If  $n$  is reasonably large, the normal quantile will be an excellent approximation.



# Statistical Properties of the OLS Estimates

- Hypothesis tests for  $\hat{\beta}_1$ :

We can test the hypothesis  $\beta_1 = 0$  against alternatives such as  $\beta_1 \neq 0$ ,  $\beta_1 > 0$ , and  $\beta_1 < 0$ .

For example, suppose we are testing the 2-sided alternative  $\beta_1 \neq 0$ . A suitable test statistic would be

$$T = \frac{\hat{\beta}_1 \hat{\sigma}_X \sqrt{n-1}}{\hat{\sigma}},$$

which has a  $t_{n-2}$  distribution (which may be approximated with a normal distribution if  $n$  is not too small).



# Confidence Interval for the Regression Line

- The fitted value at  $X$ , denoted  $\hat{Y}$ , is the  $Y$  coordinate of the estimated regression line at  $X$ :

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

The fitted value is an estimate of the regression function  $E[Y|X]$  evaluated at the point  $X$ , so we may also write  $\hat{E}[Y|X]$ .

Fitted values may be calculated at any  $X$  value. If  $X$  is one of the observed  $X$  values, say  $X = X_i$ , write  $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ .

# Confidence Interval for the Regression Line

- Since  $\hat{Y}_i$  is a random variable, we can calculate its mean and variance.  
To get the mean, recall that  $E[\hat{\beta}_0] = \beta_0$  and  $E[\hat{\beta}_1] = \beta_1$ . Therefore,

$$\begin{aligned} E[\hat{Y}_i] &= E[\hat{\beta}_0 + \hat{\beta}_1 X_i] \\ &= E[\hat{\beta}_0] + E[\hat{\beta}_1] \cdot X_i \\ &= \beta_0 + \beta_1 X_i \\ &= E[Y_i] \end{aligned}$$

Thus  $\hat{Y}_i$  is an unbiased estimate of  $E[Y|X]$  evaluated at  $X = X_i$ .

- The variance of the  $\hat{Y}_i$  is

$$\text{var}(\hat{Y}_i) = \sigma^2 \sigma_i^2$$

where

$$\sigma_i = \frac{1}{n} \left( 1 + \left( \frac{X_i - \bar{X}}{\sigma_X} \right)^2 \right)$$

# Confidence Interval for the Regression Line

- We now know the mean and variance of  $\hat{Y}_i$ . Standardizing yields

$$P \left( -1.96 \leq \frac{\hat{Y}_i - (\beta_0 + \beta_1 X_i)}{\sigma \sigma_i} \leq 1.96 \right) = .95,$$

equivalently

$$P \left( \hat{Y}_i - 1.96\sigma\sigma_i \leq \beta_0 + \beta_1 X_i \leq \hat{Y}_i + 1.96\sigma\sigma_i \right) = .95$$

This gives a **95% CI** for  $E[Y_i]$ .

# Confidence intervals for the regression line

- Since  $\sigma$  is unknown we must plug-in  $\hat{\sigma}$  for  $\sigma$  in the CI. Thus we get the approximate CI

$$P\left(\hat{Y}_i - 1.96\hat{\sigma}\sigma_i \leq \beta_0 + \beta_1 X_i \leq \hat{Y}_i + 1.96\hat{\sigma}\sigma_i\right) \approx .95$$

We can make the coverage probability exactly 0.95 by using the  $t_{n-2}$  distribution to calculate quantiles:

$$P\left(\hat{Y}_i - Q(0.975)\hat{\sigma}\sigma_i \leq \beta_0 + \beta_1 X_i \leq \hat{Y}_i + Q(0.975)\hat{\sigma}\sigma_i\right) \approx .95$$

The goal of each CI is to cover the green line, and this should happen 95% of the time.

# SLR Summary

- SLR is to estimate the population relationship with one single independent variable

$$E[Y|X = x] = \beta_0 + \beta_1 x$$

- The estimated relationship based on a particular sample is written as

$$\hat{E}[Y|X = x] = \hat{\beta}_0 + \hat{\beta}_1 x$$

- To estimate  $\beta_0$  and  $\beta_1$ , observed  $x_i$  and  $y_i$  are modeled as

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

with  $\epsilon_i$  characterized by one parameter  $\sigma$ .

- The fitted values are

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

# SLR Summary

- The OLS solution to the SLR problem is

$$\hat{\beta}_1 = \frac{\text{cov}(X, Y)}{\text{var}(X)}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\sigma} = \sqrt{\frac{\sum_i r_i^2}{n - 2}}$$

- It is not enough to only provide  $\hat{\beta}_0$  and  $\hat{\beta}_1$ . We also need to know how good these estimates are. This requires us to know the mean and variances of  $\hat{\beta}_0$  and  $\hat{\beta}_1$ .

# SLR Summary

- Both  $\hat{\beta}_0$  and  $\hat{\beta}_1$  are unbiased.

$$E[\hat{\beta}_0] = \beta_0, \quad E[\hat{\beta}_1] = \beta_1$$

- Their variances are

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{(n-1)\text{var}(X)}$$

$$\text{var}(\hat{\beta}_0) = \sigma^2 \frac{\sum x_i^2 / n}{(n-1)\text{var}(X)} = \text{var}(\hat{\beta}_1) \frac{\sum x_i^2}{n}$$



# SLR Summary

- Confidence intervals are

$$CI(\hat{\beta}_1) = \left[ \hat{\beta}_1 - c \cdot \text{SD}(\hat{\beta}_1), \hat{\beta}_1 + c \cdot \text{SD}(\hat{\beta}_1) \right]$$

$$CI(\hat{\beta}_0) = \left[ \hat{\beta}_0 - c \cdot \text{SD}(\hat{\beta}_1), \hat{\beta}_0 + c \cdot \text{SD}(\hat{\beta}_1) \right]$$

where  $c$  is the quantile of the  $t_{n-2}$  distribution for a specific confidence interval.  $n$  is the number of pairs of  $(x_i, y_i)$ . When  $n$  is large enough, usually  $\geq 30$ , the  $t_{n-2}$  distribution can be replaced with the standard normal distribution.



# SLR Summary

- $\hat{Y}_i$  is an unbiased estimate of  $E[Y|X]$  evaluated at  $X = X_i$ .

$$E[\hat{Y}_i] = E[Y_i]$$

- The variance of the  $\hat{Y}_i$  is

$$\text{var}(\hat{Y}_i) = \sigma^2 \sigma_i^2$$

where

$$\sigma_i = \frac{1}{n} \left( 1 + \left( \frac{X_i - \bar{X}}{\sigma_X} \right)^2 \right)$$

- The confidence interval of the regression line can be calculated as follows

$$CI(\hat{Y}_i) = \left[ \hat{Y}_i - c \cdot \text{SD}(\hat{Y}_i), \hat{Y}_i + c \cdot \text{SD}(\hat{Y}_i) \right]$$

where  $c$  is the quantile of the  $t_{n-2}$  distribution for a specific confidence interval.

# SLR Summary

- *Coefficient of determination*  $R^2$  is defined as follows

$$R^2 = \frac{SS_{reg}}{SS_{tot}} = 1 - \frac{SS_{err}}{SS_{tot}} = (\widehat{cor}(X, Y))^2.$$

$R^2$  tells us how much variability in the  $Y$ 's is explained by the regression. With least squares regression,  $R^2$  is also equal to the square of the correlation coefficient between the response and explanatory variables.

# Numerical Solution to SLR

- The OLS solution has an analytical solution.
- However, many machine problems do not have an analytical solution. Instead, we have to find the solution numerically.
- The SLR can be solved numerically as well and we will explain how.
- This will lay the foundation for us to find solutions to logistic regression and neural network problems later on.

**Some of the materials from this slide and onward are taken from the Machine Learning Course by Andrew Ng on Coursera.com.**

# Numerical Solution to SLR

- Notations

1.  $m$ : total number of samples in the training data

2.  $(x^{(i)}, y^{(i)})$ : the  $i$ th sample

3.  $\theta = [\theta_0, \theta_1]$

4.  $h_\theta(x) = \theta_0 + \theta_1 x$ : the affine function to be learned from the training data

- Goal: find  $\theta_0$  and  $\theta_1$  such that the following cost function is minimized

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left( h_\theta \left( x^{(i)} \right) - y^{(i)} \right)^2$$

where  $J(\theta_0, \theta_1)$  is a squared error function.

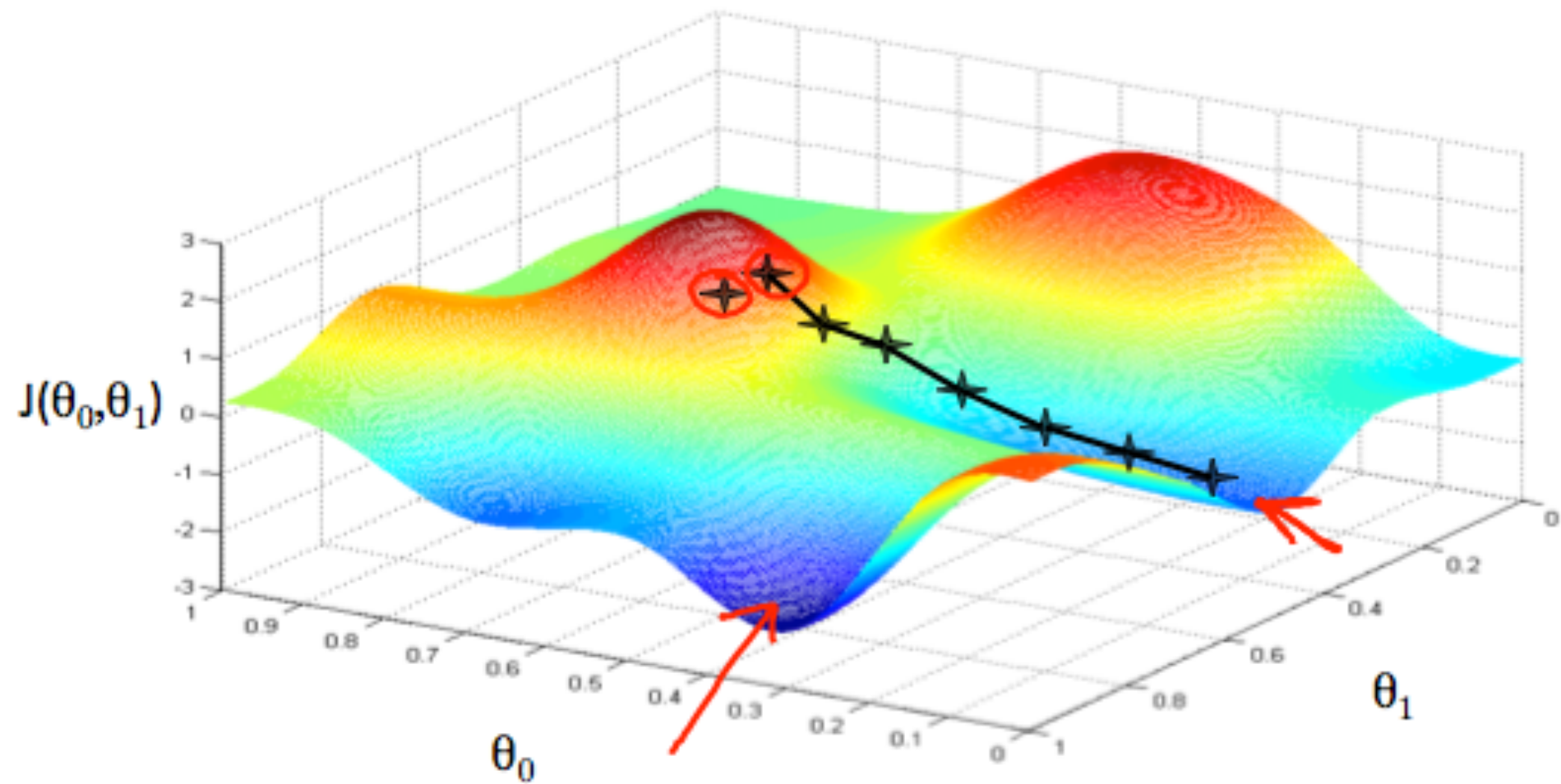
# Numerical Solution to SLR

- Procedure to find the optimal  $\theta_0$  and  $\theta_1$ :
  1. Start with initial  $\theta_0$  and  $\theta_1$ ;
  2. Keep updating  $\theta_0$  and  $\theta_1$  to reduce  $J(\theta_0, \theta_1)$  until we hopefully end up at a minimum.
- Gradient descent algorithm to update  $\theta_0$  and  $\theta_1$ :

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) \quad \text{for } j = 0 \text{ and } j = 1$$

where  $\alpha$  is called the *learning rate*.

# Numerical Solution to SLR





# Numerical Solution to SLR

- Simultaneous update

1. Correct

$$\text{temp\_0} := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\text{temp\_1} := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_0 := \text{temp\_0}$$

$$\theta_1 := \text{temp\_1}$$

2. Incorrect update

$$\text{temp\_0} := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_0 := \text{temp\_0}$$

$$\text{temp\_1} := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_1 := \text{temp\_1}$$



# Gradient descent intuition

- When the cost  $J$  is a function of only one parameter  $\theta_1$ :
  1. When the slope of  $J(\theta_1)$  is negative, the value of  $\theta_1$  increases and the gradient descent gradually takes  $\theta_1$  to the optimal value.
  2. When the slope of  $J(\theta_1)$  is positive, the value of  $\theta_1$  decreases and the gradient descent gradually takes  $\theta_1$  to the optimal value.
- Effect of  $\alpha$  on convergence:
  1. If  $\alpha$  is too small, gradient descent can be very slow.
  2. If  $\alpha$  is too large, gradient descent can overshoot the minimum. It may fail to converge, or even diverge.
- Gradient descent can converge to a local minimum, even with a fixed learning rate  $\alpha$ . As we approach a local minimum, gradient descent will automatically take smaller steps.

# Gradient descent for SLR

- With cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left( h_{\theta} \left( x^{(i)} \right) - y^{(i)} \right)^2 = \frac{1}{2m} \sum_{i=1}^m \left( \theta_0 + \theta_1 x^{(i)} - y^{(i)} \right)^2$$

where  $m$  is the size of the training set and  $x^{(i)}, y^{(i)}$  are the values of the training set data.

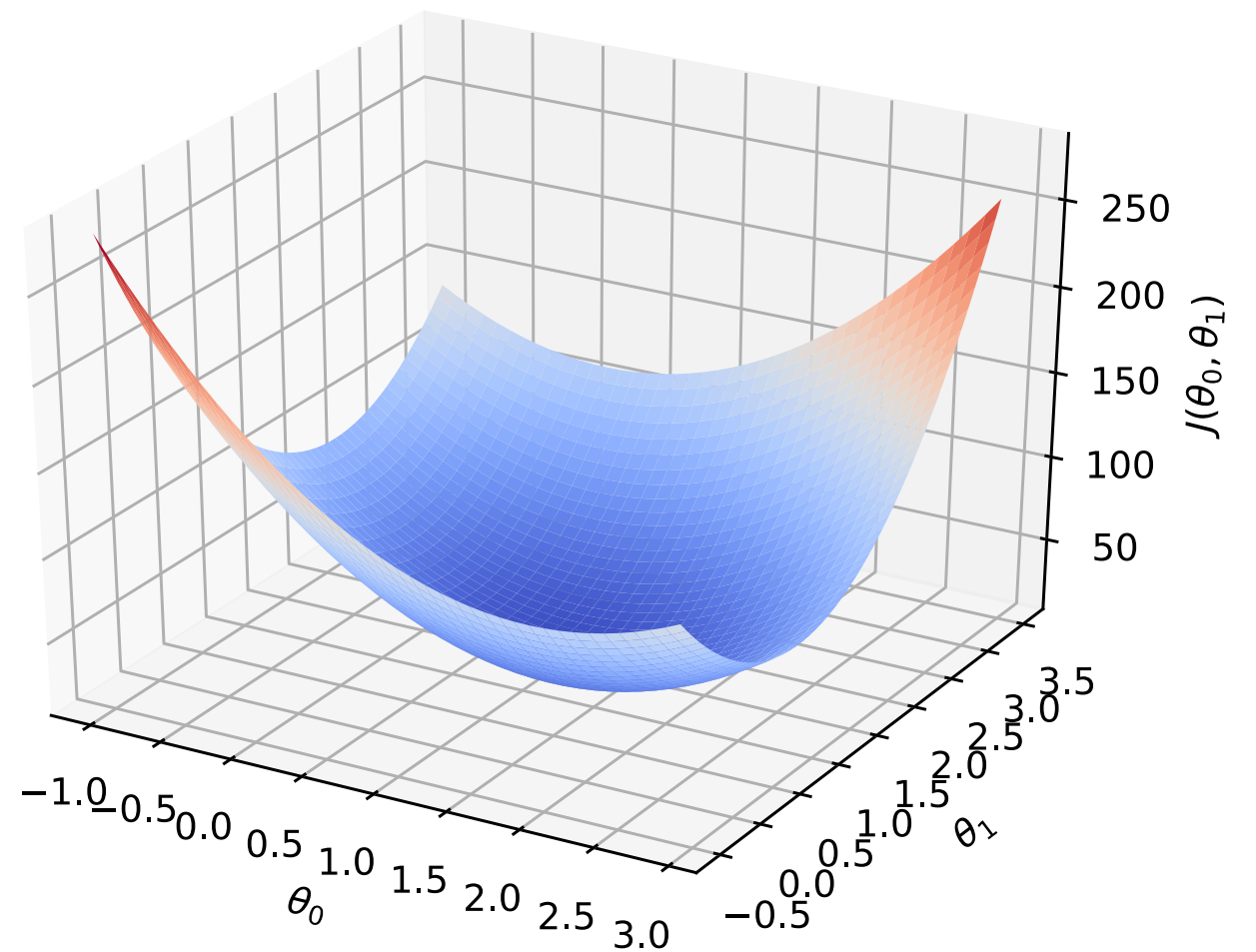
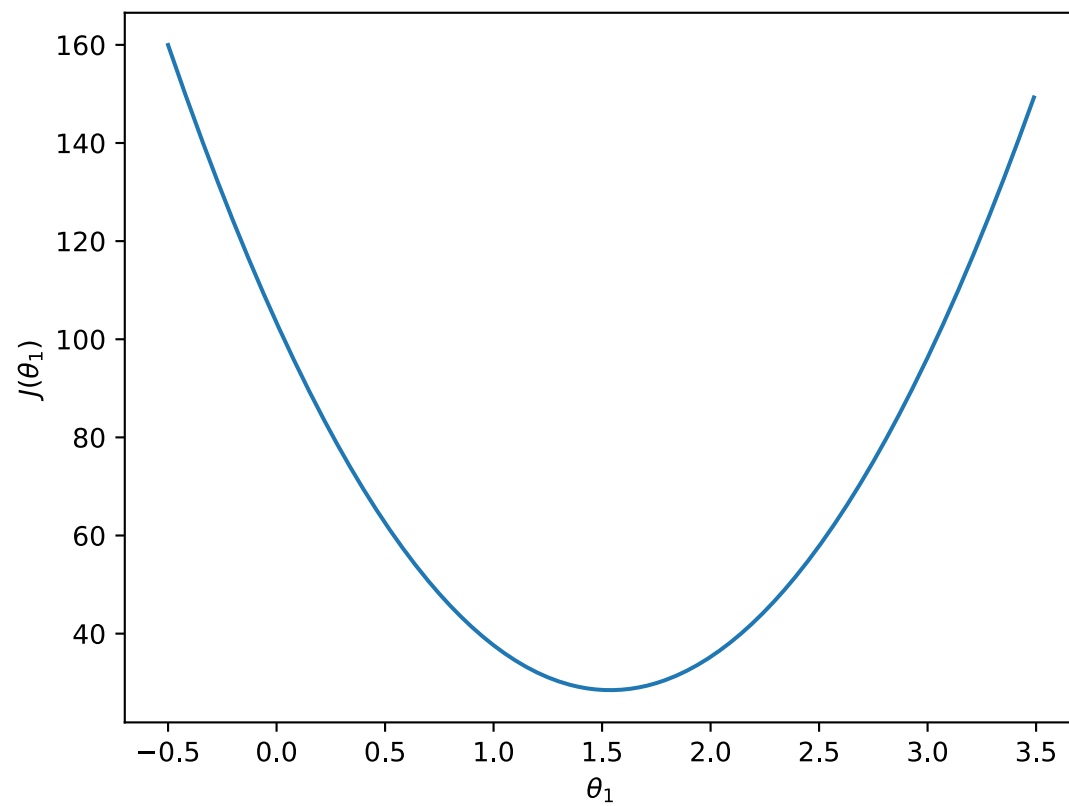
- We can get the gradient as follows:

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m \left( h_{\theta} \left( x^{(i)} \right) - y^{(i)} \right) = \frac{1}{m} \sum_{i=1}^m \left( \theta_0 + \theta_1 x^{(i)} - y^{(i)} \right)$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m \left( h_{\theta} \left( x^{(i)} \right) - y^{(i)} \right) x^{(i)} = \frac{1}{m} \sum_{i=1}^m \left( \theta_0 + \theta_1 x^{(i)} - y^{(i)} \right) x^{(i)}$$

- For every step of the iteration, the data in the entire training set is used. This is called *batch gradient descent*.

# Gradient descent for SLR



# Multivariate Linear Regression

# Multiple variables and notation

| size | number of bedrooms | number of floors | age of home | price (\$1000) |
|------|--------------------|------------------|-------------|----------------|
| 2104 | 5                  | 2                | 34          | 450            |
| 2500 | 4                  | 2                | 23          | 320            |
| 743  | 2                  | 1                | 43          | 120            |
| 1500 | 3                  | 1                | 21          | 200            |
| ...  | ...                | ...              | ...         | ...            |

- **Notation:**

$n$  = number of features (variables)

$x^{(i)}$  = input (features) of the  $i^{th}$  training example.

$x_j^{(i)}$  = value of feature  $j$  in the  $i^{th}$  training example.

- **Hypothesis:**

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

# Multiple variables and notation

- In general:

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

- For convenience of notation, define  $x_0 = 1$ . Then

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix}$$

and

$$h_{\theta}(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n = \theta^T x$$

# Cost function

- Cost function:

$$J(\theta_0, \theta_1, \dots, \theta_n) = J(\theta) = \frac{1}{2m} \sum_{i=1}^m \left( h_{\theta} \left( x^{(i)} \right) - y^{(i)} \right)^2$$



# Gradient descent

- Simultaneously update parameters based on gradient descent:

$$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m \left( h_{\theta} \left( x^{(i)} \right) - y^{(i)} \right) x_j^{(i)} \quad \text{for } j = 0, 1, \dots, n$$

That is:

$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m \left( h_{\theta} \left( x^{(i)} \right) - y^{(i)} \right) x_0^{(i)}$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m \left( h_{\theta} \left( x^{(i)} \right) - y^{(i)} \right) x_1^{(i)}$$

⋮

$$\theta_n := \theta_n - \alpha \frac{1}{m} \sum_{i=1}^m \left( h_{\theta} \left( x^{(i)} \right) - y^{(i)} \right) x_n^{(i)}$$



# Gradient descent in practice

- Scale the features by doing one of the following:
  1. Get every feature into approximately the  $-1 \leq x_i \leq 1$  range.
  2. Make features have zero mean by

$$x_i := \frac{x_i - \mu_i}{\text{range of } x_i}$$

# Gradient descent in practice

- Make sure gradient descent is working correctly by plotting the cost function on the y axis and the number of iterations on the x axis. If  $J(\theta)$  ever increases, then you probably need to decrease  $\alpha$ .
- Declare convergence if  $J(\theta)$  decreases by less than  $E$  in one iteration, where  $E$  is some small value such as  $10^{-3}$ . However in practice it's difficult to choose this threshold value.
- For sufficiently small  $\alpha$ ,  $J(\theta)$  should decrease on every iteration. But if  $\alpha$  is too small, gradient descent can be slow to converge.

# Introduction

- A regression analysis estimates the *conditional expectations* of the dependent variable  $Y$  given the independent variables  $X$  (Do NOT be confused with independent random variables) – that is, the average value of the dependent variable when the independent variables are held fixed.

$$E[Y|X = x] = f(x, \beta)$$

where the function  $f$  is called the *regression function* and  $\beta$  denotes the unknown parameters.  $\beta$  can be a scalar or a vector.

$X$  is also called the *predictor variable*, *explanatory variable*, *regressor*, *covariates*, or *input variable*.

$Y$  is also called regressand or response variable.